

Physical exercise intervention in depressive disorders: meta-analysis and systematic review.

Previous meta-analyses investigating the effect of exercise on depression have included trials where the control condition has been categorized as placebo despite the fact that this particular placebo intervention (e.g., meditation, relaxation) has been recognized as having an antidepressant effect. Because meditation and mindfulness-based interventions are associated with depression reduction, it is impossible to separate the effect of the physical exercise from the meditation-related parts. The present study determined the efficacy of exercise in reducing symptoms of depression compared with no treatment, placebo conditions or usual care among clinically defined depressed adults. Of 89 retrieved studies, 15 passed the inclusion criteria of which 13 studies presented sufficient information for calculating effect sizes. The main result showed a significant large overall effect favoring exercise intervention. The effect size was even larger when only trials that had used no treatment or placebo conditions were analyzed. Nevertheless, effect size was reduced to a moderate level when only studies with high methodological quality were included in the analysis. Exercise may be recommended for people with mild and moderate depression who are willing, motivated, and physically healthy enough to engage in such a program. © 2013 John Wiley & Sons A/S. Published by John Wiley & Sons Ltd.